
EECS 16A Imaging 3

****TA, UCS1, UCS1, UCS1****

Announcements

- Next week (the week of 10/02) is buffer:
 - You can make up **one** missed lab from the Imaging Module, if needed (unless you have received approval to make-up multiple labs).
 - Fill out the sign-up form to attend:
 - The form will be released on Ed in the weekly post.

Last time: Matrix-vector multiplication

1	0	0	0	0	0	0	0	...
0	1	0	0	0	0	0	0	...
0	0	1	0	0	0	0	0	...
0	0	0	1	0	0	0	0	...
0	0	0	0	1	0	0	0	...
0	0	0	0	0	1	0	0	...
0	0	0	0	0	0	1	0	...
0	0	0	0	0	0	0	1	...
...								

Masking Matrix H

i_1
i_2
i_3
i_n

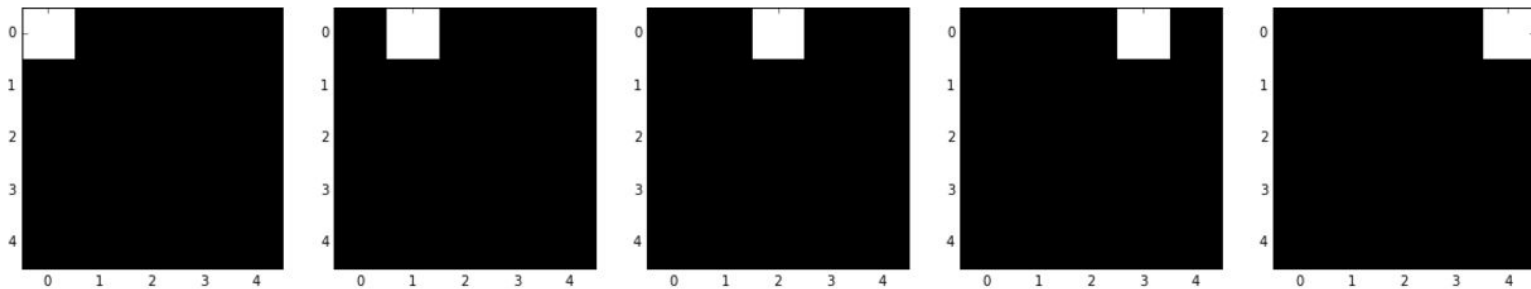
Unknown,
vectorized
image, $\vec{i}$

=

s_1
s_2
s_3
s_n

Recorded
Sensor
readings, $\vec{s}$

Last time: Single-pixel scanning



- Setup a masking matrix where each row is a mask
 - Measured each pixel individually once

$$\vec{s} = H\vec{i}$$

- How did we reconstruct our image, once we had s ?

Poll Time! (this is review)

What are the requirements of our masking matrix H ?
(multiple choice)

- A. H is invertible
- B. H has linearly independent columns
- C. H has a trivial nullspace
- D. Determinant of H is 0
- E. Unique solution to $H\vec{i} = \vec{s}$

$$\vec{s} = H\vec{i}$$

Our system

Poll Time! (this is review)

What are the requirements of our masking matrix H ?
(multiple choice)

- A. H is invertible
- B. H has linearly independent columns
- C. H has a trivial nullspace
- D. Determinant of H is 0
- E. Unique solution to $H\vec{i} = \vec{s}$

$$\vec{s} = H\vec{i}$$

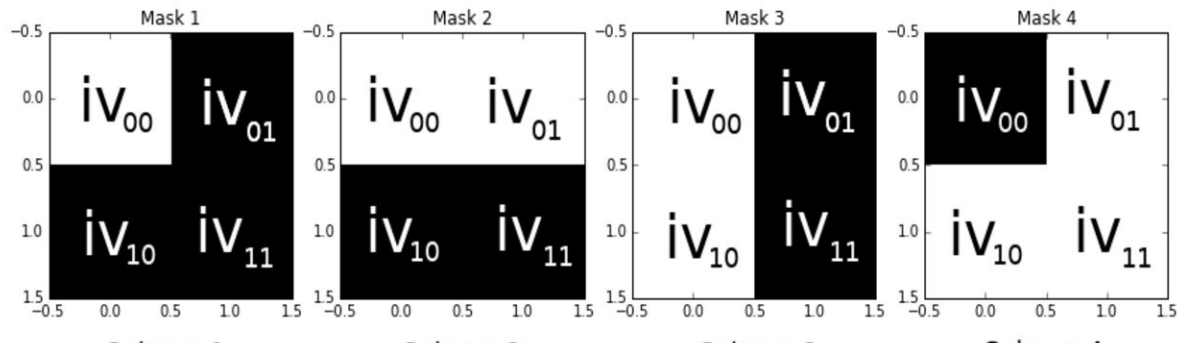
Our system

Questions from Imaging 2

Goal: Understand which measurements are good measurements

- ✓ Can we always reconstruct our image? **Need invertible H**
- ? Are all invertible matrices equally good as scanning matrices?
- ? What happens if we mess up a single scan?
- ? What if we use multiple pixel instead of single pixel scan?

Today: Multipixel scanning



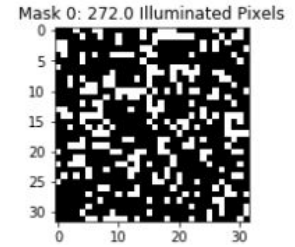
- **Can we measure multiple pixels at a time?**
 - Measurements are now linear combinations of pixels
- **How can we reconstruct our scanned image?**
 - Can multipixel masks still be linearly independent, aka invertible?

Why do we care?

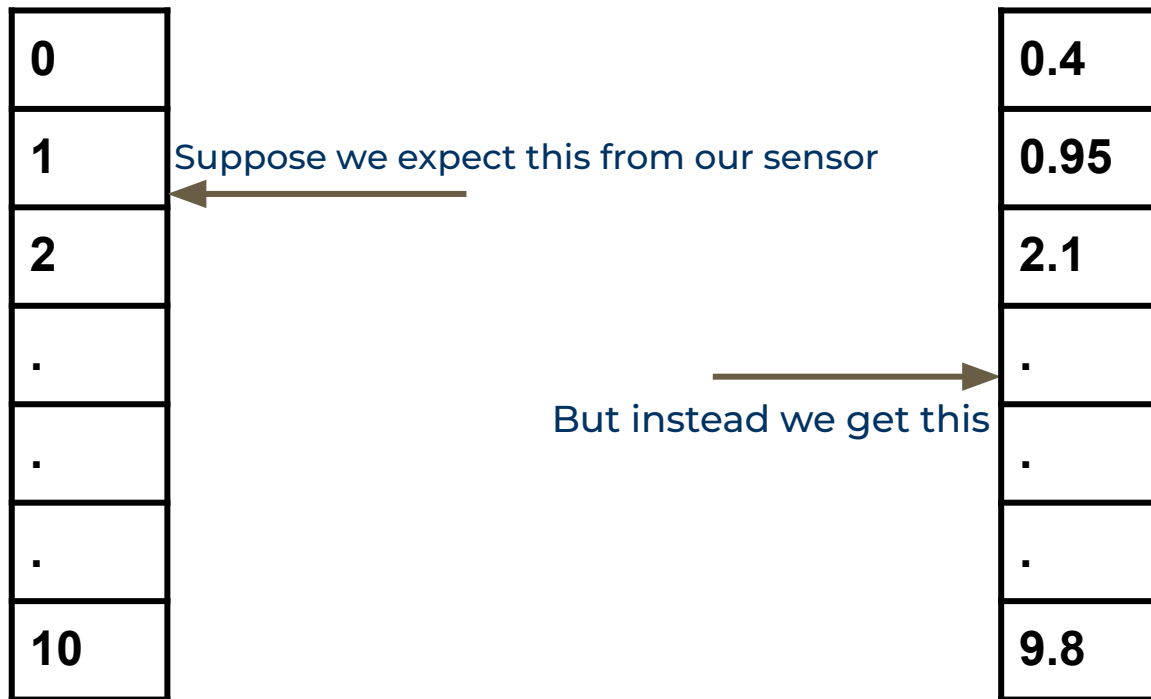
- Improve image quality by **redundancy**:
 - Ideally, one measurement is enough
 - Redundancy: conducting more measurements
 - Extract pixel value by averaging over multiple measurements
 - Good measurements → good average
 - Occasional bad measurements → worsen the pixel value but makes it tolerant of some errors

How do we do it?

- Illuminate multiple pixels per scan:
 - Each mask measures a linear combination of pixels instead of a single pixel, i.e. has multiple 1's
 - Increases signal level:
 - Signal is data that we do want (ex: light intensity from pixel illumination)
- Problems:
 - Our measurements are noisy
 - **Noise** is a random unwanted variation in our measurement (ex: room light getting into box)
 - Noise may be amplified through inverting a matrix!
- **Goal: high signal, low noise** → high signal-to-noise ratio (SNR)



What is noise?



What is noise?

0.4
0.95
2.1
.
.
.
9.8

$\vec{s}_{real}$

Measured values =
ideal vector + noise vector (ω)

=

0
1
2
.
.
.
10

$\vec{s}_{ideal}$

+

0.4
-0.05
0.1
.
.
.
-0.2

$\vec{\omega}$

How does noise affect our system?

1	0	0	0	0	0	0	0	...
0	1	0	0	0	0	0	0	...
0	0	1	0	0	0	0	0	...
0	0	0	1	0	0	0	0	...
0	0	0	0	1	0	0	0	...
0	0	0	0	0	1	0	0	...
0	0	0	0	0	0	1	0	...
0	0	0	0	0	0	0	1	...
...								

Masking Matrix H

i_1
i_2
i_3
i_n

Unknown,
vectorized
image, $\vec{i}$

+

ω_1
ω_2
ω_3
ω_n

Random
noise
vector, $\vec{w}$

=

s_1
s_2
s_3
s_n

Recorded
Sensor
readings, $\vec{s}$

A more realistic system

- Sensor readings = H applied to image vector + noise vector

$$\vec{s} = H\vec{i} + \vec{w}$$

- We can't reconstruct exact $\vec{i}$, but we can estimate it

$$\vec{i}_{est} = H^{-1}\vec{s} = \vec{i} + \boxed{H^{-1}\vec{w}}$$

**Be careful about the reconstruction error (noise term)
or else it could blow up !!**

Eigenvalues for inverse matrices

- H is an invertible NxN matrix $\leftrightarrow$ trivial nullspace
 - No zero eigenvalues
- Assume H has N linearly independent eigenvectors
 - $Hv_i = \lambda_i v_i$ for $i = 1 \dots N$
- They span $\mathbb{R}^N \rightarrow$ they span the noise vector!
- Eigenvalue-eigenvector definition:

$$H^{-1}v_i = \frac{1}{\lambda_i}v_i \text{ for } i = 1 \dots N$$

Proof:

$$\begin{aligned} H\vec{v}_i &= \lambda_i \vec{v}_i \\ \implies H^{-1}H\vec{v}_i &= \lambda_i H^{-1}\vec{v}_i \\ \implies H^{-1}\vec{v}_i &= \frac{1}{\lambda_i}\vec{v}_i \end{aligned}$$

How do eigenvalues affect noise?

The noise vector can be written as a linear combination of eigenvectors:

$$\vec{\omega} = \alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \cdots \alpha_n \vec{v}_n$$

Including effect of H^{-1}

$$H^{-1} \vec{\omega} = H^{-1} (\alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \cdots \alpha_n \vec{v}_n)$$

Rewritten with eigenvalues:

$$H^{-1} \vec{\omega} = \frac{1}{\lambda_1} \alpha_1 \vec{v}_1 + \frac{1}{\lambda_2} \alpha_2 \vec{v}_2 + \cdots \frac{1}{\lambda_n} \alpha_n \vec{v}_n$$

Linking it all together

$$\vec{i}_{est} = H^{-1}\vec{s} = \vec{i} + \boxed{H^{-1}\vec{w}}$$
$$\boxed{H^{-1}\vec{w}} = \frac{1}{\lambda_1}\alpha_1\vec{v}_1 + \frac{1}{\lambda_2}\alpha_2\vec{v}_2 + \cdots \frac{1}{\lambda_n}\alpha_n\vec{v}_n$$

- Remember: want small noise term for high signal-to-noise ratio
- The noise is directly related to the eigenvalues

Poll Time!

- Do we want small or large eigenvalues for the H matrix in order to get a good image?
 - A. Large
 - B. The magnitude doesn't matter
 - C. Small
- Which of the following equations correctly model our imaging system? (multiple choice)
 - A. $s_{\text{ideal}} = H.i$
 - B. $s_{\text{real}} = s_{\text{ideal}} + w = H.i + w$
 - C. $s_{\text{real}} = s_{\text{ideal}} + w = H.i + H.w$
 - D. $i_{\text{est}} = H^{-1}.s_{\text{real}} = H^{-1}.s_{\text{ideal}} + H^{-1}.w$
 - E. $i_{\text{est}} = H^{-1}.s_{\text{real}} = H^{-1}.s_{\text{ideal}} + w$

Poll Time!

- Do we want small or large eigenvalues for the H matrix in order to get a good image?

A. Large

B. The magnitude doesn't matter

C. Small

- Which of the following equations correctly model our imaging system? (multiple choice)

A. $s_{\text{ideal}} = H.i$

B. $s_{\text{real}} = s_{\text{ideal}} + w = H.i + w$

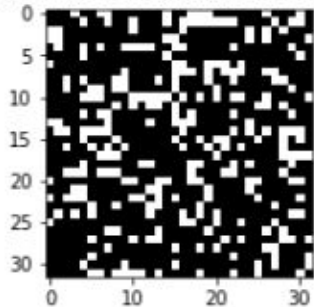
C. $s_{\text{real}} = s_{\text{ideal}} + w = H.i + H.w$

D. $i_{\text{est}} = H^{-1}.s_{\text{real}} = H^{-1}.s_{\text{ideal}} + H^{-1}.w$

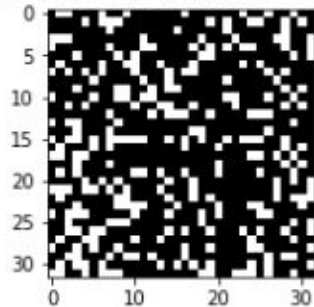
E. $i_{\text{est}} = H^{-1}.s_{\text{real}} = H^{-1}.s_{\text{ideal}} + w$

Possible scanning matrix: Random

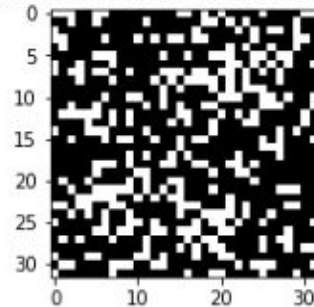
Mask 0: 272.0 Illuminated Pixels



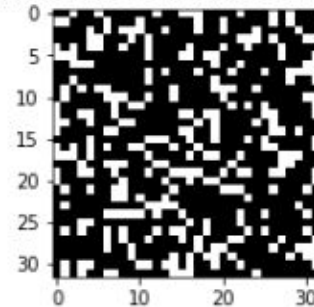
Mask 1: 281.0 Illuminated Pixels



Mask 2: 313.0 Illuminated Pixels



Mask 3: 289.0 Illuminated Pixels

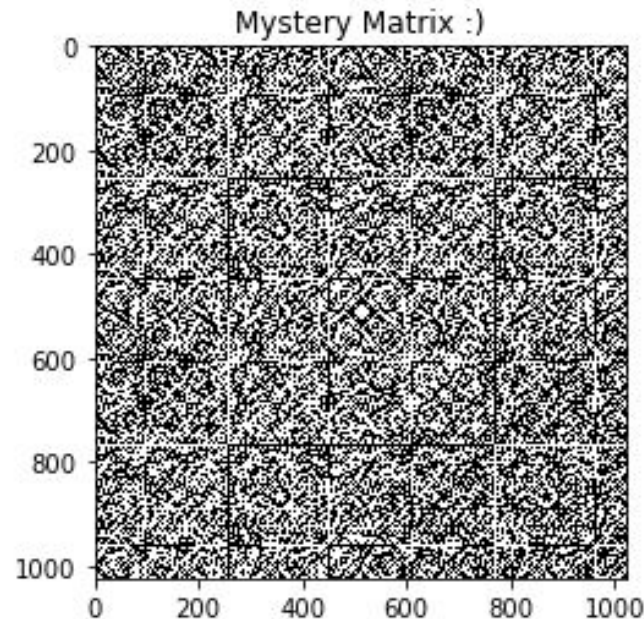


- Illuminate ~300 pixels per scan
 - *Usually* invertible
 - But what are its eigenvalues?

╰(ツ)╯

A more systematic scanning matrix

- Hadamard matrix!
- Constructed to have large eigenvalues
 - Just what we need!



Multipixel Scanning Use Cases

- Not the “superior technology” – as any practice, it has its advantages and disadvantages:
 - Multipixel scan is useful if we cared about getting *close* to each pixel value, prioritizing getting decent results for each pixel
 - Single-pixel scan allows to get really good measurements for some pixels while other pixels will be lost entirely

Example: Horsetail Falls



Do you see
the missing
pixel?

Example: Horsetail Falls Comparison



Single-pixel: many “perfect” pixels but some are entirely missing



Multipixel: fewer “perfect” pixels (creating blurriness) but information-preserving

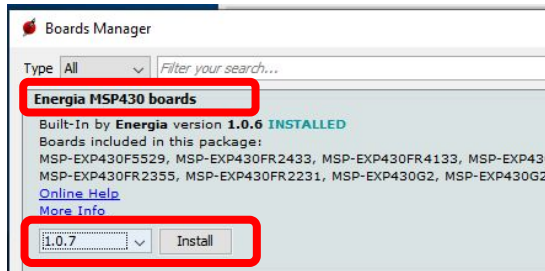
Multipixel Scanning Expectations

- Note: when shining light at multiple portions of an image, our light will easily bleed to the pixels around one region
 - *Therefore, we may not get better results*
- **Today's goal:**
 - Show that it's possible to get results using multipixel scanning
 - **Use case of multipixel scanning at the end of the lab**

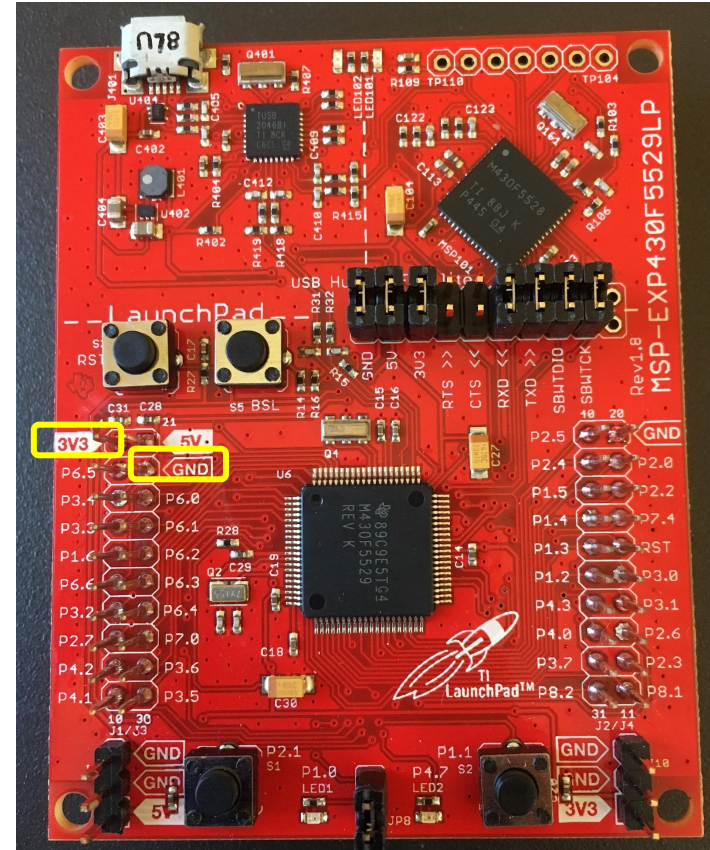
LaunchPad



- Today's microcontroller
- Use Energia to upload code
 - If error occurs, go to **Tools** → **Board** → **Board Manager**, and install 1.0.7 board version:



- Check COM port number
- Same loan/return policy



Workflow and Debugging

- READ CAREFULLY
- Circuit:
 - Resistor is different
 - Check light sensor orientation
 - **Red** jumper for **+**
 - **Black** jumper for **GND**
- Projector:
 - **BE CAREFUL WITH PORTS (DC, HDMI)**
 - brightness 0, contrast 100
 - may restart in the middle of scan
- Image: not-too-detailed
- Cover box with jacket for dark conditions
- Project masks onto image
- Reconstruct image: $H^{-1}\mathbf{s}_{\text{real}} = \mathbf{i}_{\text{est}}$

Checkoff Form:



Feedback Form:

